

Ethan H Reeves

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Education

University of California, Santa Barbara

B.S. in Electrical Computer Engineering – College of Engineering

Sep 2023 - June 2027 (Expected)

GPA: 3.76 / 4.00

Relevant Coursework

- High Speed IC Design (M.S.)
- Nanoelectronic Device Physics
- VLSI Design I & II*
- Computer Architecture I & II
- Integrated Circuit Fabrication (Labs in Cleanroom)
- Data Structures & Algorithms
- Operating Systems*
- Sensor / Peripheral Design*

*Courses in progress

Experience

Engineering Intern at *Spin Master Bay Area Innovation Center* (“BAIC”) | Pleasanton, CA

Aug - Sep 2025

- Assembled controllers used by design engineers to test the mechanical operations in electronic toys
- Produced prototypes including writing embedded software; optimized designs by reducing components used and cost inputs

Electrical Engineering Undergraduate Researcher at *Universidad Carlos III de Madrid* | Madrid, Spain

Jun - Aug 2025

- Collaborated in an international research environment to develop real-time testing protocols for the power grid
- Conducted simulations using an RTDS Simulator to model electric grid behavior under various fault events
- Researched Grid Forming versus Grid Following inverter technologies for enhanced grid stability and fault tolerance

Projects

Fabricated Working NMOS from Silicon Wafer | *Semiconductor Fabrication, Electrical Testing*

Jan - Mar 2026

- Fabricated a working NMOS, Resistor, and Capacitor from scratch in the UCSB Engineering II Cleanroom
- Performed all fabrication steps, including masking, metrology, doping, etching, oxide growth, & metal deposition

Research Paper on Cu vs Graphene Interconnects | *Research*

Jan - Mar 2026

- Developed resistivity / resistance models for copper and ruthenium interconnects to analyze nanowire and temperature effects
- Simulated monolayer graphene in Synopsys QuantumATK to calculate resistivity at finite temperature for comparison

Working RISC-V Pipeline Processor in Verilog | *Computer Architecture, Design, Testing*

Jan - Mar 2026

- Processor supported add, addi, sub, and, andi, or, slt, slti, lw, sw, beq, jal, & lui instructions
- Implemented a controller, branch predictor, and critical word first / prefetch cache system to reduce execution time

Parallel Prefix Adder in Cadence Virtuoso | *Design, Testing, Validation*

December 2025

- Designed a functional 500 MHz 16-bit parallel prefix adder in Cadence Virtuoso
- Developed a circuit design with custom gate logic on the Skywater 130nm PDK

Layout of CMOS Logic Gates in Cadence Virtuoso | *Layout, Design, Testing, Validation*

Sep - Dec 2025

- Designed functional INV, NAND, NOR, MUX, and XOR gates using the Skywater 130nm PDK
- Tested and verified layout designs using DRC and LVS rules

Goleta Sound Machine | *Design, Testing*

Oct - Dec 2023

- Developed a guitar tuner and effects unit using a Teensy microcontroller and audio adapter board
- Implemented real-time frequency analysis, LCD output, and programmable audio effects

Technical Skills

EDA Tools: Cadence Virtuoso, Logism, Synopsys HSPICE, Quartus II, Synopsys QuantumATK, Siemens ModelSim

Coding Languages Assembly (6502 / RISC-V), C, C++, Java, Python, SPICE, Verilog / SystemVerilog

Technical Skills Chip fabrication, circuit design, CMOS Design, FPGA Development, Fusion 360, Git, Microscope Soldering, Oscilloscope, Raspberry Pi, RSCAD FX, RTDS Simulator, Teensy

Extracurriculars

UCSB Theta Tau - Professional Co-Ed Engineering Fraternity *President*

Jun 2025 - Present

UCSB TASA (Taiwanese American Student Association) *Co-Historian (Photographer)*

Jun 2025 - Present

Hobbies - Guitar, Photography ([@ethan_r_photo](#)), Piano (Liszt, Debussy), Reading (Blake Crouch), Soccer, Surfing, and Swimming